

Z-TransformScript: Chapter 9 (z -transform, transfer functions of discrete-time systems)**Learning goals**

- Compute z -transforms with their region of convergence (ROC) and invert rational $X(z)$ by partial fractions for causal, anticausal and two-sided sequences.
- Derive $H(z)$, poles and zeros of the RPM smoothers of Sheet 3 and decide stability from the pole positions.
- Read the magnitude response from the pole-zero map (geometric interpretation) and design a two-pole resonator for the knock frequency.
- Build a pole-zero explorer (polynomial roots, ASCII pole-zero map, frequency and impulse response).
- Implement the resonator in Q15 fixed point on the Arduino Uno and observe coefficient quantisation, zero-input limit cycles and overflow oscillations for $r \rightarrow 1$.

Background

The (bilateral) z -transform of $x[n]$ is $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$, defined in a ring-shaped ROC $r_1 < |z| < r_2$. An LTI system with the difference equation $\sum_k a_k y[n-k] = \sum_k b_k x[n-k]$ has the transfer function

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \dots + a_N z^{-N}} = \frac{b_0}{a_0} z^{N-M} \frac{\prod_k (z - z_k)}{\prod_k (z - p_k)}.$$

A causal system is stable iff all poles lie inside the unit circle (then the ROC $|z| > \max |p_k|$ contains the unit circle), and its frequency response is $H(e^{j\Omega})$, $\Omega = 2\pi f/f_s$. Geometrically, $|H(e^{j\Omega})| = |\frac{b_0}{a_0}| \prod_k |e^{j\Omega} - z_k| / \prod_k |e^{j\Omega} - p_k|$: a pole close to the unit circle produces a peak, a zero on it a notch.

The **two-pole resonator** with poles $p_{1,2} = re^{\pm j\theta}$, $\theta = 2\pi f_0/f_s$,

$$H(z) = \frac{b_0}{1 + a_1 z^{-1} + a_2 z^{-2}}, \quad a_1 = -2r \cos \theta, \quad a_2 = r^2, \quad (1)$$

is the simplest digital knock filter. Throughout this sheet $f_0 = 6.5$ kHz and $f_s = 40$ kHz ($\theta = 58.5^\circ$).

Exercise 9.1 Analytical z -transforms**PAPER**

- Compute the z -transform and the ROC of $\delta[n-k]$, $\varepsilon[n]$, $a^n \varepsilon[n]$, $-a^n \varepsilon[-n-1]$ and $r^n \cos(\theta n) \varepsilon[n]$. Why is the ROC indispensable?
- Find all sequences with $X(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 0.8z^{-1})}$ by partial fractions (one for each possible ROC). Which one is causal, which one has a DTFT? Check $x[0]$ and $x[1]$ of the causal one with the difference equation.
- RPM smoothers of Sheet 3.** Derive $H(z)$, poles and zeros of
 - the exponential smoother $y[n] = a y[n-1] + (1-a)x[n]$,
 - the M -point moving average.

When are they stable? Give $|H|$ at $\Omega = 0$ and $\Omega = \pi$ for $a = 0.9$ and the notch frequencies of the moving average for $M = 8$ at $f_s = 40$ kHz. What happens to the pole at $z = 1$ of $(1 - z^{-M})/(1 - z^{-1})$?

- d) **Resonator.** Show that (1) has the poles $re^{\pm j\theta}$. Derive the stability triangle in the (a_1, a_2) -plane ($|a_2| < 1$, $|a_1| < 1 + a_2$) and mark the resonator.
- e) Using the geometric interpretation, show that the -3 dB bandwidth is $\Delta\Omega \approx 2(1 - r)$, i.e. $B \approx (1 - r)f_s/\pi$, for r close to 1. Choose r for $B = 400$ Hz. Compare with the impulse-invariant image $r = e^{-T/\tau}$ of the analog knock resonator of Sheet 7 ($\tau = 0.8$ ms).
- f) Determine b_0 such that $|H(e^{j\theta})| = 1$ and evaluate it for $r = 0.97$.

Exercise 9.2 Pole-zero explorer

JSLINUX

Program `code/jslinux/sheet09/pz_explorer.c` reads the coefficients from the command line,

```
1 ./pz_explorer -fs 40000 -b 0.1 -a 1 -0.9      # exponential smoother
2 ./pz_explorer -fs 40000 -res 6500 0.97        # resonator, gain 1 at f0
```

computes poles and zeros, prints them in polar form (radius, angle in Hz), decides stability, draws the pole-zero map with `ap_pz()` and plots magnitude (dB) and phase over f and the impulse response.

- a) Poles and zeros are the roots of $z^L B(z)$ and $z^L A(z)$, $L = \max(M, N)$ (`tf_roots()` is given). Implement the Durand–Kerner iteration `poly_roots()` for a polynomial of degree n with $c_0 = 1$:

$$r_i \leftarrow r_i - \frac{c(r_i)}{\prod_{j \neq i} (r_i - r_j)}, \quad r_i^{(0)} = (0.4 + 0.9j)^{i+1}.$$

Use the small complex struct of the template (no `complex.h`).

- b) Implement the geometric evaluation `freq_geom()` and the impulse response from the difference equation. The program compares the geometric magnitude with the direct evaluation of B/A .
- c) Explore and report (pole-zero map, peak/notch, bandwidth, $|h[n]|$):
- exponential smoother $a = 0.9$; moving average $M = 8$;
 - resonator $r = 0.9, 0.97, 0.99$: compare the -3 dB bandwidth with $(1 - r)f_s/\pi$;
 - resonator $r = 1$ and $r = 1.02$: what does the program say, what does the impulse response do?
 - a notch filter at 6.5 kHz with zeros at $e^{\pm j\theta}$ and poles at $0.95e^{\pm j\theta}$. Give the coefficients and the notch width; why is $|H| \approx 1$ far from the notch?
- d) Why may the magnitude plot of an unstable system be misleading?

Exercise 9.3 Digital resonator in float and Q15

WOKWI JSLINUX

The resonator (1) is implemented twice in the header `resonator.h`: in `float` and in fixed point with signals in Q15 (`int16_t`, 1 LSB = 2^{-15}), coefficients in Q14 and an `int32_t` accumulator. The *same file* is used on JSLinux and on the Uno; you find it in both folders `code/jslinux/sheet09/` and `code/wokwi/sheet09_resonator/`.

- a) (**paper**) Why can a_1 not be stored in Q15 here? In which format is the accumulator, and how many bits does it need at most? Which quantisation step does $a_2 = r^2$ get, and how large is the resulting change of the pole radius for $r = 0.9999$?
- b) Implement the functions `res_design`, `res_f_step` and `res_q_step` (rounding or truncation, saturation or wrap-around, and count the overflows). Note: `int` has 16 bits on the AVR – cast to `int32_t` *before* multiplying. Run the program `limitcycle.c` and compare quantised and exact pole positions (part (a) of the program) and the impulse responses (b).

- c) **Zero-input limit cycles.** Starting from $y[-1] = 0.5$ with $x = 0$, the float resonator decays to zero. Part (c) of the program measures the amplitude of the Q15 output after 5 s. For rounding, a limit cycle can persist as long as rounding maps $a_2 y$ back to y : derive the dead-band bound $|y| \leq \frac{0.5}{1-a_2}$ LSB. Compare with the measured amplitudes for $r \rightarrow 1$, rounding and truncation, and explain the frequency of the limit cycle.
- d) **Overflow.** Without normalisation ($b_0 = 1$) the gain at f_0 is about 20 (Exercise 9.1 f). Part (d) feeds a knock burst of amplitude $A = 0.1$ and 0.5 into the resonator with $r = 0.99$. Compare wrap-around and saturation. What is an *overflow oscillation*?
- e) **Uno.** Open the Wokwi project (Uno, potentiometer = r from 0.90 to 1.00, switch SW1 = rounding/truncation, SW2 = saturation/wrap, button = next test T0/T1/T2, see the sketch header) with your `resonator.h`. Reproduce the results of c) and d) for $r \approx 0.97, 0.99, 0.999$. Are the integer results identical to JSLinux? Could you also expect identical float results?
- f) Draw conclusions for a fixed-point knock filter in an ECU (choice of r , rounding, overflow handling, scaling).

Hint: The Uno computes the test signals itself; its ADC (about 9 kHz maximum, Sheet 5) could not sample a 6.5 kHz knock signal anyway. The tests are not real-time – we study the arithmetic; real-time knock filtering follows on Sheet 10.

Deliverables

Solutions of 9.1 (partial fractions with all ROCs, stability triangle, bandwidth formula and r for 400 Hz); the pole-zero explorer with the screenshots (ASCII plots) and the result table of 9.2 c); the tables of 9.3 a), c) and d) from JSLinux and the corresponding Uno outputs (T0–T2 for at least two values of r) with your conclusions 9.3 f).